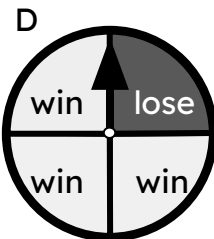
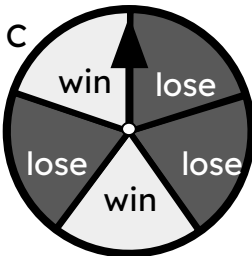
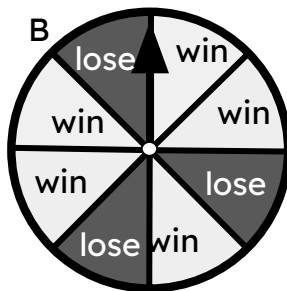
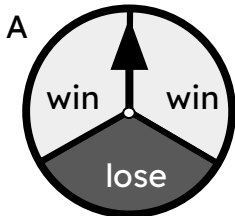




Calculating theoretical probabilities from lists (one event)

Task A: Finding probabilities of events with spinners

1) Here are four win/lose spinners.



a) For each spinner, find the probability that it lands on 'win' in a single spin.

b) Sofia wants to choose the spinner that gives her the best chance of winning. Which spinner should she choose?

c) Which spinner is least likely to land on win?

2) Alex, Jun and Sam spin the spinner 40 times in total.
They get a point every time the spinner lands on their name.

a) Who should expect to get the most points?

b) Who should expect to get the fewest points?

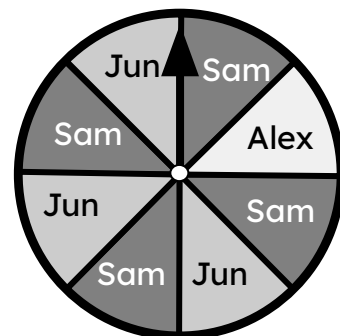
c) What is the probability the spinner lands on Alex?

d) What is the probability the spinner lands on Jun?

e) What is the probability the spinner lands on Sam?

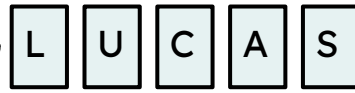
f) How many more points should Sam expect to get compared to Alex?

g) Will your answer to (f) necessarily happen when they spin the spinner? Explain your answer.



Task B: Finding probabilities of events

- 1) Andeep and Lucas each have a deck of cards spelling out their own name. They play a game where they each randomly select a card from their deck, record the letter and then return the card to the deck.



They win a point if they draw a card containing the letter A.

- a) What are the probabilities for each player winning a point?
- b) Who is most likely to get a point when they select a card? *Lucas*
- c) They change the rules so that they get a point if they select a vowel.
Who is most likely to get a point with these rules? Explain your answer.

- 2) A regular six-sided dice is rolled. Find the following probabilities.

- a) $P(3) =$ b) $P(4) =$ c) $P(\text{not } 4) =$
- d) $P(\text{even}) =$ e) $P(\text{odd}) =$ f) $P(\text{multiple of } 3) =$
- g) $P(7) =$ h) $P(\text{not } 7) =$ i) $P(\text{factor of } 60) =$
- j) The dice is rolled 60 times. Approximately how many times could it be expected to land on the number 3, based on its probability?
- k) Will your answer to part (j) necessarily happen? Explain your answer.